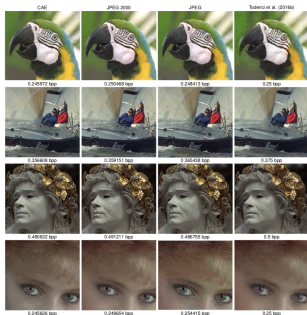


# Deep Learning

## Specific Autoencoders and Applications



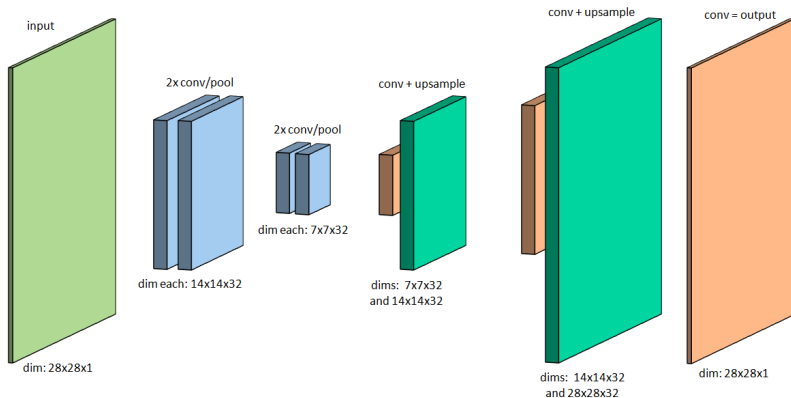
### Learning goals

- convolutional AEs
- applications of AEs

# CONVOLUTIONAL AUTOENCODER (CONVAE)

- For the image domain, using convolutions is advantageous. Can we also make use of them in AEs?
- In a ConvAE, the encoder consists of convolutional layers. The decoder, on the other hand, consists of transpose convolution layers or simple upsampling operations.

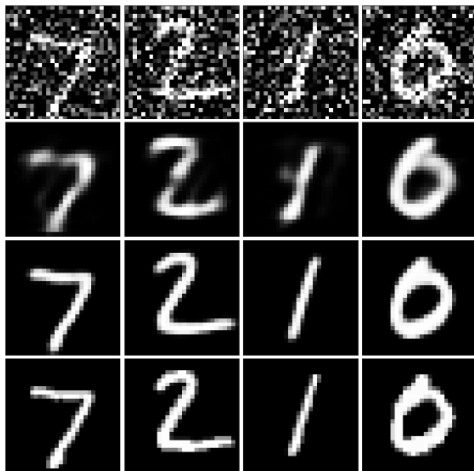
# CONVOLUTIONAL AUTOENCODER (CONVAE)



**Figure:** Potential architecture of a convolutional autoencoder.

We now apply this architecture to denoise MNIST.

# CONVOLUTIONAL AUTOENCODER (CONVAE)



**Figure:** Top row: noised data, second row: AE with  $\dim(\mathbf{z}) = 32$  (roughly 50k params), third row: ConvAE (roughly 25k params), fourth row: ground truth.

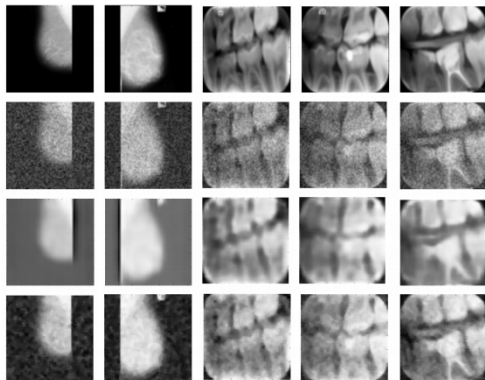
# REAL-WORLD APPLICATIONS

Today, autoencoders are still used for tasks such as:

- data de-noising,
- compression,
- and dimensionality reduction for the purpose of visualization.

# REAL-WORLD APPLICATIONS

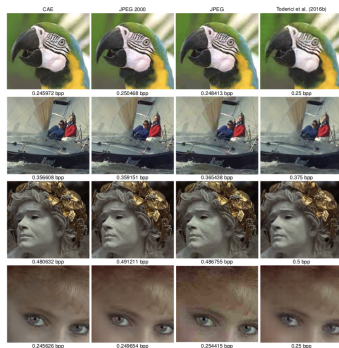
**Medical image denoising** using convolutional denoising autoencoders.



**Figure:** Top row: real image, second row: noisy version, third row: results of a (convolutional) denoising autoencoder and fourth row: results of a median filter (Gondara, 2016).

# REAL-WORLD APPLICATIONS

## AE-based image compression.



**Figure:** Closeups of images produced by different compression algorithms (Theis et al., 2017).

# REFERENCES



Theis, L., Shi, W., Cunningham, A., & Huszár, F. (2017). *Lossy Image Compression with Compressive Autoencoders*.



Gondara, L. (2016). Medical Image Denoising Using Convolutional Denoising Autoencoders. *2016 IEEE 16th International Conference on Data Mining Workshops (ICDMW)*, 241–246. <https://doi.org/10.1109/ICDMW.2016.0041>